Yunpeng Wang

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Areas of focus: Robotics, Machine Learning, Distributed Systems

Education

University of Pennsylvania *Aug 2022–May 2025*
M.S.E., Electrical Engineering; M.S.E., Computer and Information Science | GPA: 3.70 Philadelphia, PA

Central South University *Sep 2018–Jun 2022*
B.Eng., Internet of Things Engineering | GPA: 3.77 Changsha, China

Professional Experience

WeRide *Jul 2025–Present*
Software Engineer San Jose, CA

- Maintain and extend planning-simulation and scheduling systems used for continuous autonomous-driving evaluation.
- Develop incident-analysis workflows using vision-language models and decision-tree classifiers.

WeRide *Jun 2024–Dec 2024*
Software Engineer Intern San Jose, CA

- Built a microservice-based simulation launcher for continuous evaluation and testing of autonomous-driving models.
- Improved bag-decoding throughput 40%, cut memory use 42%, and resolved 50+ Sentry-tracked stability issues.

Alibaba Cloud *Jun 2023–Sep 2023*
Software Engineering Intern Beijing, China

- Built an AIOps storage alerting service using 10+ anomaly detectors, including Local Outlier Factor.
- Integrated model configuration, visualization, alert management, and notifications into an operations dashboard.

Research Experience

xLab, GRASP Laboratory, University of Pennsylvania *Jan 2025–Present*
Research Assistant [code] Philadelphia, PA

- Developed and evaluated a F1TENTH sim-to-real reinforcement-learning pipeline using SAC, PPO, recurrent policies, Pure Pursuit demonstrations, F1TENTH Gym, Stable-Baselines3, and a ROS deployment node.
- Randomized vehicle dynamics, LiDAR and state noise, and command latency across parallel simulation environments; modeled dropped, repeated, and delayed commands explicitly.
- Deployed selected policies to physical hardware. An early SAC policy recorded approximately 16-second simulation and 30-second hardware laps; a later imitation-initialized policy reached approximately seven seconds in simulation but transferred poorly because of aggressive control.
- Prototyped parameter-conditioned and recurrent control, an LSTM parameter estimator, and a decentralized two-car interface; contextual control and overtaking remained incomplete.

Central South University *Jun 2020–Feb 2021*
Research Assistant Changsha, China

- Developed a vision pipeline for reading pointer-type water meters under varying illumination and camera poses.
- Used MSER and NMS for contours, SURF voting for alignment, and neighbor-based angle correction.

Robotics and Autonomous Systems Projects

Robot Arm Perception and Manipulation for Block Stacking

- Contributed the path-sequencing framework, reusable robot configurations, camera–end-effector–base transformations, and initial static-block orientation logic.
- Used damped least-squares inverse kinematics near a singular grasp and a wait-and-pick strategy for moving blocks.
- The system stacked four static and three moving blocks in a five-minute hardware run, placing first in class.

Drone Path Planning, Trajectory Generation, and Control

- Combined 26-connected A* search, map inflation, path simplification, trajectory generation, and geometric control.
- Implemented and timed fifth-order minimum-jerk trajectory segments for the visual-inertial extension.
- Three Vicon-guided maze flights kept team-reported error below 0.1 m on most paths and 0.2 m near turns.

Extended and Unscented Kalman Filters

- Implemented an augmented-state EKF that estimated a simulated nonlinear-system parameter from an initial value of -10 toward its true value of -1 over 100 observations.
- Implemented a manifold-consistent quaternion UKF for IMU orientation and angular-velocity estimation, calibrated sensor axes, and compared the resulting trajectories qualitatively with Vicon measurements.

LiDAR Particle-Filter SLAM

- Implemented 1,000 pose hypotheses, odometry proposals, scan-to-map scoring, log-sum-exp weight normalization, effective-sample-size-triggered stratified resampling, and log-odds occupancy updates at 0.05 m resolution.
- Processed four recorded datasets and produced occupancy maps and trajectory overlays; evaluation remained qualitative because no independent ground-truth trajectory was available.

NeRF Implementation

- Combined COLMAP poses, ray sampling, sine coordinate features, a three-layer MLP, and differentiable compositing.
- Trained on 32×32 images for 2,000 iterations and evaluated same-view reconstructions qualitatively.
- Documented deviations from canonical NeRF, including omitted view direction and cosine features, unconstrained color and density, and a nonstandard inclusive transmittance calculation.

Proximal Policy Optimization for Planar Walker Control

- Adapted a PyTorch PPO baseline to DeepMind Control with environment conversion, generalized advantage estimation, minibatching, learning-rate annealing, gradient clipping, KL early stopping, checkpoints, and diagnostics.
- Trained a policy that produced walking behavior in simulation; retained a single-run entropy comparison as an implementation diagnostic rather than a causal ablation.

Machine Learning Projects

Deterministic and Stochastic Binary Weight Quantization [code]

- Compared deterministic and stochastic binary quantization while retaining full-precision training weights.
- In the recorded runs, deterministic $\{-1, +1\}$ binarization achieved the highest observed validation accuracy; deterministic $\{0, 1\}$ reached approximately 76% and produced a zero-dominant weight distribution.

An Evaluation of Positional-Embedding Integration in Transformers [code]

- Derived additive positional encoding as a constrained form of concatenation with shared projection blocks and evaluated four integration variants on Multi30k translation.
- Found no quality gain from concatenation; doubling embedding width raised mean epoch time from 55.10 to 138.53 s.
- Conducted a separate periodic-signal reconstruction experiment that found no systematic relationship between signal frequency and the tested positional-frequency range.

Thesis Quality Prediction

- Developed a pipeline using BERT representations and an LSTM to estimate essay quality from users' writing behavior; dataset, authorship, and evaluation details remain to be recovered.

Distributed Systems and Software Projects

FinanceTracker: VLM-Assisted Bookkeeping Platform

- Built a FastAPI/RQ pipeline for CSV, XLSX, and PDF ingestion with a PostgreSQL-backed Next.js UI.
- Used Qwen-based structured inference and OCR for table detection, schema mapping, categorization, duplicate grouping, and counteraction review while retaining source history and audit events.
- Packaged the frontend, API, worker, Redis, PostgreSQL, MinIO, and local VLM with Docker Compose; deployed release v1.1 to a home server and documented a successful six-file acceptance import.

CSPR: Cross-Source Personal Recommendations

- Built ten Typer command-line tools for workspace management, browser-history and bookmark evidence, preference and read memory, RSS collection, item merging, source inference, feedback, and Markdown newspaper output.
- Implemented URL canonicalization, persistent seen-item filtering, local/private-host exclusion, resumable history scans, source-frequency inference, and cross-input deduplication.
- Packaged the Python wheel with four agent skills and 85 automated tests covering command-line and data workflows.

Distributed Web Search Engine

- Built a custom HTTP/1.1 server, persistent distributed key-value store, RDD-style processing layer, crawler, inverted index, iterative PageRank computation, and online query service.
- Integrated storage, distributed processing, indexing, ranking, and a query interface into a functional demonstration.
- Used a TF-IDF-like score whose inverse-frequency numerator was vocabulary size rather than conventional corpus document count; project artifacts did not preserve a controlled scale or retrieval-quality evaluation.

Raft Consensus [code]

- Implemented leader election, log replication, persistence, snapshot compaction, and `InstallSnapshot` recovery.
- Used mutex-based concurrency control and stress tests to diagnose races, deadlocks, and election-timer behavior.

Chord Distributed Hash Table

- Implemented joining, stabilization, finger tables, successor lookup, failure detection, and a transaction-ID callback registry for asynchronous RPC-style continuations.

JOS Operating System Implementation [code]

- Implemented assigned components for physical and virtual memory, user environments and traps, copy-on-write fork, preemptive round-robin scheduling, and interprocess communication.
- Added a file server, block cache, and E1000 network driver using memory-mapped I/O and DMA descriptor rings.

Oat Compiler Implementation

- Implemented an x86lite simulator, LLVMlite-to-x86-64 backend, lexer and parser, Oat-to-LLVM frontend, and type checker supporting structs, function pointers, and subtyping.
- Built a generic worklist dataflow solver plus liveness, alias, and constant-propagation analyses; used the results for dead-code elimination and iterative optimization passes.
- Implemented greedy liveness-aware allocation and interference-graph coloring with precolored registers and spilling.

Course Review Web Application

- Built and deployed a Django application on Amazon EC2, then reimplemented it with Spring Boot and Angular.
- Implemented course creation, update, deletion, and multi-field search across course metadata.

Smart Home System

- The team system combined Raspberry Pi face-based access control, STM32F429 touchscreen control for curtains and lighting, and a proposed robot-following health-monitoring module with YOLO-based fall detection.

Patents and Software Registrations

- **Chinese patent application 202111412755.9.** Wireless weighing system for vending machines; status unverified.
- **Chinese software copyright registration 2021SR1099107.** “Software for Reading Pointer-Type Water Meters.”

Honors and Awards

- First place among 150 teams, Hunan Collegiate Programming Contest, 2021.
- Meritorious Winner, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling, 2021.
- First place, MEAM 5200 robotic block-stacking class competition, 2023 (team result).

Technical Skills

- **Programming languages:** Python, C++, C, Java, Go, OCaml, Bash, SQL, MATLAB, TypeScript.
- **Robotics:** ROS/ROS 2, F1TENTH Gym, DeepMind Control, MuJoCo, Vicon, AprilTag, LiDAR, COLMAP, UE5.
- **Machine learning and data:** PyTorch, Transformers, Stable-Baselines3, NumPy, pandas, scikit-learn, Matplotlib.
- **Systems and cloud:** Git, Docker, AWS EC2, Kubernetes, Bazel, Helm, ns-3, PostgreSQL, Redis, MinIO, Sentry.
- **Web development:** FastAPI, Next.js, Django, Flask, SQLAlchemy, Spring Boot, Angular, Vue.